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Multi-scale method for ill-conditioned linear systems

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Abstract. A multi-scale method is presented for solving ill-conditioned linear systems of equations. Numerical examples are provided to show the effectiveness and efficiency of the proposed method compared with traditional iterative methods.

1. Introduction

The linear system of equations

$$Ax = b, \tag{1}$$

where $A = (a_{ij}) \in \Re^{n\times n}$ is nonsingular, $x = (x_j) \in \Re^n$ unknown and $b \in \Re^n$ known, plays an important role in science and engineering computing [1]. There are two approaches for solving (1) — direct methods and iterative methods [2]. For large sparse systems (1), the iterative methods are popular and efficient. Many iterative methods have been presented, such as Jacobi, Gauss-Seidel, SOR, Conjugate Gradient, Subspace, Preconditioned, GMRES methods [3]. However, it's difficult to develop effective methods for ill-conditioned linear systems whose coefficient matrices have large condition numbers [4-22].

In this paper a multi-scale method is suggested to solve ill-conditioned system (1). Through the numerical examples it is showed that the new method is effective and more efficient than the others.

The paper is organized as follows. First, the method is presented. Then, numerical analysis is conducted. Finally, the numerical examples are provided and the conclusion is made.

2. The multi-scale method

If the coefficient matrix A of the equations (1) is ill-conditioned, the system (1) can be solved by the following equations

$$\widetilde{A}\widetilde{x} = b, \tag{2}$$

where $\widetilde{A}$ and $\widetilde{x}$ are as follows

$$\widetilde{A} = \begin{pmatrix} \frac{1}{\|\alpha_1\|_2} a_{11} & \frac{1}{\|\alpha_2\|_2} a_{12} & \cdots & \frac{1}{\|\alpha_n\|_2} a_{1n} \\ \frac{1}{\|\alpha_1\|_2} a_{21} & \frac{1}{\|\alpha_2\|_2} a_{22} & \cdots & \frac{1}{\|\alpha_n\|_2} a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{1}{\|\alpha_1\|_2} a_{n1} & \frac{1}{\|\alpha_2\|_2} a_{n2} & \cdots & \frac{1}{\|\alpha_n\|_2} a_{nn} \end{pmatrix}, \quad \widetilde{x} = \begin{pmatrix} \|\alpha_1\|_2 x_1 \\ \|\alpha_2\|_2 x_2 \\ \vdots \\ \|\alpha_n\|_2 x_n \end{pmatrix},$$

here α_j, $j = 1,2,\cdots,n$ is the jth column of A.

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With (2), the following multi-scale method is obtained for system (1).

THE MULTI-SCALE ALGORITHM

$Step\,1$ solve $\widetilde{A}\widetilde{x} = b$ with iterative methods;

$$Step\,2 \quad x = \begin{pmatrix} \frac{1}{\|\alpha_1\|_2}\widetilde{x}_1 \\ \frac{1}{\|\alpha_2\|_2}\widetilde{x}_2 \\ \vdots \\ \frac{1}{\|\alpha_n\|_2}\widetilde{x}_n \end{pmatrix}.$$

The following remarks show that the multi-scale method could improve the condition number of the coefficient matrix.

Remark 1. From (2), it yields

$$\widetilde{A} = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{n1} & a_{n2} & \cdots & a_{nn} \end{pmatrix} \begin{pmatrix} \frac{1}{\|\alpha_1\|_2} & 0 & \cdots & 0 \\ 0 & \frac{1}{\|\alpha_2\|_2} & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \frac{1}{\|\alpha_n\|_2} \end{pmatrix} \overset{\Delta}{=} AD.$$

Thus, for the condition number with1-norm, denoted by

$$d_{\max} = \max\{\|\alpha_1\|_2, \|\alpha_2\|_2, \cdots, \|\alpha_n\|_2\},\ d_{\min} = \min\{\|\alpha_1\|_2, \|\alpha_2\|_2, \cdots, \|\alpha_n\|_2\},$$

the following holds

$$\begin{aligned} \operatorname{cond}(\widetilde{A}) &= \|\widetilde{A}\|\|\widetilde{A}^{-1}\| = \|AD\|\|D^{-1}A^{-1}\| \\ &\le \|A\|\|D\|\|D^{-1}\|\|A^{-1}\| = \tfrac{d_{\max}}{d_{\min}}\|A\|\|A^{-1}\| = \tfrac{d_{\max}}{d_{\min}}\operatorname{cond}(A). \end{aligned} \tag{3}$$

Therefore, if $d_{\min}$ is very close to $d_{\max}$, then the condition number cond($\widetilde{A}$) could be no more than the condition number cond(A).

Remark 2. If D is taken as an orthogonal matrix, then the following can be obtained from (3)

$$\operatorname{cond}(\widetilde{A}) \le \operatorname{cond}(A).$$

3. Numerical examples

In this section, numerical examples are given to illustrate the multi-scale method. The initial guess is 0 and the stopping criterion is

$$\frac{\|r^{(k)}\|_2}{\|r^{(0)}\|_2} < 10^{-6},$$

where $r^{(k)}$ is the residual vector after k iterations. The results are compared with classical iterative methods.

Example 1. Consider the following system

$$\begin{pmatrix} 1 & 1 & 0 \\ 0 & 100 & 1 \\ 1 & 0 & 10000 \end{pmatrix} \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 2 \\ 101 \\ 10001 \end{pmatrix}.$$

The coefficient matrix A is almost singular. The condition number of A is 1.0×10^4 ($\|A\|_2\|A^{-1}\|_2$). The exact solution is $(1, 1, 1)^T$. The condition number of $\widetilde{A}$ is 2.4. The numerical results are listed in Table 1 compared with classical Jacobi and Gauss-Seidel methods.

Table 1. Iterations (IT) and relative error (ERR) for Example 1.

Jacobi		Gauss-Seidel		Multi-scale	
IT	ERR	IT	ERR	IT	ERR
5	5.8e-9	4	5.8e-9	6	1.0e-12

Example 2. Consider the following system

$$\begin{pmatrix} 1 & \frac{1}{2} & \frac{1}{3} \\ \frac{1}{2} & \frac{1}{3} & \frac{1}{4} \\ \frac{1}{3} & \frac{1}{4} & \frac{1}{5} \end{pmatrix}\begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix}.$$

The coefficient matrix A is almost singular. The condition number of A is 524.1 ($\|A\|_2\|A^{-1}\|_2$). The exact solution is $(3, -24, 30)^T$. The condition number of $\widetilde{A}$ is 373.7. The numerical results are listed in Table 2 compared with classical Jacobi and Gauss-Seidel methods.

Table 2. Iterations (IT) and relative error (ERR) for Example 2.

Jacobi		Gauss-Seidel		Multi-scale	
IT	ERR	IT	ERR	IT	ERR
1305	NaN	907	2.6e-8	880	4.4e-8

4. Conclusions

In this paper, a multi-scale method is presented to solve system of linear equations (1) which is ill-conditioned. The method improves the condition number of the ill-conditioned system. Numerical results show that the multi-scale method is effective and efficient for linear ill-conditioned problems.

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